PRECALCULUS · VECTORS

Vector Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Vector Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. If $|u| = 3$ and $|v| = 4$, can $|u + v|$ be 7? Can it be 1? Can it be 8?
   7 if u and v point the same way; 1 if they point in opposite directions; never 8, because $|u + v| \le 7$.
2. Find $\langle 2, -1\rangle + \langle 3, 5\rangle$.
   $\langle 5, 4\rangle$

2 Explore use the tool

1. Make $u = \langle 4, 3\rangle$. Record its magnitude and direction.
   Magnitude 5; direction ≈ 36.9°.
2. Make $u + v = \langle 0, 0\rangle$. How are u and v related?
   v = –u: same length, opposite direction.
3. Find vectors with $|u + v| = |u| + |v|$. What must be true about u and v?
   They point the same way (v is a positive multiple of u).
4. Switch to $u - v$. Describe where $u - v$ appears on the picture.
   It goes from the tip of v to the tip of u.

3 Record use the tool and fill in the table

Operations

u	v	u + v	u − v	2u	|u + v|
⟨2, 1⟩	⟨1, 3⟩	⟨3, 4⟩	⟨1, –2⟩	⟨4, 2⟩	5
⟨–3, 2⟩	⟨3, –2⟩	⟨0, 0⟩	⟨–6, 4⟩	⟨–6, 4⟩	0
⟨4, 0⟩	⟨0, 3⟩	⟨4, 3⟩	⟨4, –3⟩	⟨8, 0⟩	5
⟨1.5, 2⟩	⟨3, 4⟩	⟨4.5, 6⟩	⟨–1.5, –2⟩	⟨3, 4⟩	7.5

4 Explain what did you discover?

1. Add vectors by adding their components.
2. $|u + v| \le$ |u| + |v|, with equality only when u and v point the same way.
3. $k \cdot u$ multiplies the length by |k|. If $k < 0$ it also reverses the direction.
4. $u - v = u +$ (–v), and it goes from the tip of v to the tip of u.
5. A plane flies with velocity $\langle 300, 0\rangle$ km/h into a wind of $\langle -40, 30\rangle$ km/h (wind speed 50 km/h). Explain why its ground speed is not 300 − 50 or 300 + 50.
   Velocities add as vectors: $\langle 260, 30\rangle$, with magnitude $\sqrt{(260^2 + 30^2)} \approx 261.7$ km/h. Magnitudes only add or subtract like that when the vectors point along the same line.

5 Apply a new situation, no tool

1. Find $|\langle -5, 12\rangle|$ and a unit vector in its direction.

   13; $\langle -\frac{5}{13}, \frac{12}{13}\rangle$

2. $u = \langle 2, -3\rangle$ and $v = \langle -1, 4\rangle$. Find $3u - 2v$.

   $\langle 8, -17\rangle$

3. Find the direction angle of $\langle -3, 3\rangle$.

   135°

4. A vector has magnitude 10 and direction 30°. Find its components.

   $\langle 5\sqrt{3}, 5\rangle \approx \langle 8.66, 5\rangle$

Teacher notes

Intended discovery: Vectors add component by component (tip to tail); magnitudes satisfy $|u + v| \le |u| + |v|$ with equality only for vectors pointing the same way; $k \cdot u$ scales length by $|k|$ and reverses direction for $k < 0$.

Likely misconceptions:

- Adding magnitudes instead of vectors.
- Drawing $u - v$ from the tip of u to the tip of v (it is the reverse).
- Using the calculator's arctan without checking the quadrant (Apply 3).

Answers that can vary: Explore 2 and 3 and the Try this list accept many vector pairs.